



The Ulan infantry fighting vehicle, on maneuvers

Key Manufacturers General Dynamics

General Dynamics is one of a new breed of highly adaptable military-industrial conglomerates. Following the end of the Cold War, its future looked bleak, but by concentrating on armored vehicles, warships, and military information systems, it was able to rise to prominence again.

IN 1982, GENERAL DYNAMICS, until then a major force in submarine construction and military aviation, took a decision to move into the manufacturing of fighting vehicles, and a new division, Land Systems, was created to acquire the Chrysler Corporation's defense interests. Its main asset was the M60 Patton Main Battle Tank, over 15,000 of which were produced between 1961 and 1987. These equipped the US Army and US Marine Corps' armored divisions throughout much of the Cold War, and saw active service (by now in its third generation) with the USMC in the 1991 Gulf War.

American fighting vehicles are often named after historic senior commanders, and the Patton's successor, the M1 Abrams, was no exception. After a lengthy design process, it entered service in 1980 and soon demonstrated its superiority. Successive upgrades enabled it to remain active for many decades. Its original composite armor was progressively improved, most significantly by adding depleted uranium or "reactive" (explosive) plates to the most vulnerable areas (see pp.238–39), while the original rifled 105 mm gun, the M68A1, deemed inadequate for the modern battlefield, was soon

Flyer Advanced Light Strike Vehicle

Developed for Special Forces, the Flyer carries up to nine men at up to 100 mph (160 km/h). It can be armed with a machine gun, a cannon, or a 40 mm grenade launcher.



replaced by the smoothbore 120 mm M256A1, designed in Germany. This 44-caliber cannon could fire a variety of projectiles, including the M829 APFSDS (Armor Piercing Fin Stabilized Discarding Sabot) "dart"—made of depleted uranium, and capable of penetrating 22 in (570 mm) of steel armor at 2,200 yds (2,000 m)—as well as high-explosive (shaped charge) and antipersonnel cartridges containing over a thousand $\frac{3}{8}$ in (9.5 mm) tungsten balls.

General Dynamics had divested itself of all of its military aviation interests by the end of the 20th century, but Land Systems soon expanded further, with acquisitions from Europe as well as at home. First came Santa Bárbara Sistemas, acquired from the Spanish Government, which produced not only vehicles but also small arms, munitions, and missiles. Next, in 2003, Land Systems acquired General Motors' defense interests, and then Steyr Daimler Puch Spezialfahrzeug (SDPS) from an Austrian investment house. The latter brought with it the Swiss MOWAG company, which had been producing specialized military and civilian vehicles with a degree of success since 1950. These new European interests soon became an important part of the parent company's armored vehicle development effort, with Santa Bárbara and Steyr working together (as ASCOD—Austrian-Spanish Cooperation Development) to produce the Pizarro Infantry Fighting Vehicle (known as the Ulan in Austrian service) and the Scout SV (Specialist Vehicle). The Pizarro/Ulan was a limited success, adopted by Austria and Spain only, but the Scout

SV was a different story. In preference to BAE Systems' CV90, it was adopted by the British Army as the Ajax family, to replace its aging Combat Vehicle Reconnaissance (Tracked) family of vehicles.

SDPS independently developed the (wheeled) Pandur Armored Fighting Vehicle from a design produced by another Spanish concern, Pegaso, while MOWAG produced the Eagle, a light tactical vehicle, the DURO, an off-road tactical transport, and, most successfully, the Piranha family of wheeled multirole APC/IFVs. The Piranha entered service in 1972, and was soon available in four distinct versions, from four- to 10-wheeled, some of them equipped with twin propellers and rudders to give them a limited "smooth-water" amphibious capability. The Piranha was to become the basis for the eight-wheeled LAV-25 and Bison, used by US and Canadian units, and the latter's six-wheeled AVGP (Armored Vehicle General Purpose), known in their various forms as Cougar, Grizzly, and Husky, as well as the eight-wheeled LAV III known as the Kodiak. Later variants of the Piranha formed the basis of the US Army's Stryker family of armored fighting vehicles, almost 4,500 of which had entered service when



Abrams production

Production of Abrams MBTs began at Detroit and the Lima plant at Ohio. When the Detroit plant closed in 1996, Lima took over refurbishment duties. The Lima plant had previously built tanks such as the Sherman.